

7(a)	(induced) e.m.f. is (directly) proportional to rate	M1
	of change of (magnetic) flux (linkage)	A1
7(b)(i)	$\Phi = BA$	C1
	$= 7.2 \times 10^{-3} \times 3.2 \times 10^{-4}$	A1
	$= 2.3 \times 10^{-6} \text{ Wb}$	
7(b)(ii)	tangent drawn at steepest point on Fig. 7.2	C1
	evidence of multiplication by 340	C1
	maximum rate of change of flux = 0.82 Wb s^{-1}	A1
7(b)(iii)	$V_0 = 0.82 \text{ V}$ or V_0 given as identical numerical answer to the answer in (b)(ii)	A1
7(b)(iv)	sinusoidal curve of period 2.0 ms from $t = 0$ to $t = 6.0 \text{ ms}$	B1
	all peaks at $+V_0$ and all troughs at $-V_0$	B1
	line showing $V = 0$ at (and only at) $t = 0, 1.0, 2.0, 3.0, 4.0, 5.0$ and 6.0 ms	B1
7(b)(v)	$A = 0.82 \text{ V}$ or A has same numerical value as answer in (b)(iii) , with unit V	A1
	$B = 2\pi / (2.0 \times 10^{-3})$	C1
	$= 3100 \text{ rad s}^{-1}$	A1